Peripheral Blood Smear White Blood Cell Classification Using Deep Learning

AI/ML Specialisation Capstone Project – Problem Statement

Field	Details
Project Number	To be assigned by the programme/mentor
Domain	Healthcare AI / Medical Image Analysis / Computer Vision
Project Type	Individual capstone project (subject to mentor approval)
Primary Dataset	White Blood Cells Dataset
Dataset Source URL	https://www.kaggle.com/datasets/masoudnickparvar/white-blood-cells-dataset/data
Core Curriculum Alignment	Exploratory Data Analysis (EDA), image preprocessing, supervised learning, CNNs, transfer learning, model evaluation, explainability/error analysis, and prototype deployment
Expected Prototype	Streamlit application for uploading an individual WBC image and viewing the predicted class and confidence score

1. Background and Context

White blood cells (WBCs) are important components of the immune system. Their identification and classification in peripheral blood smear images form part of hematological assessment. Manual morphological review requires trained personnel and may be time-consuming. Computer vision and deep-learning methods can be explored as tools to classify cropped WBC images and support research into automated hematology image analysis.

2. Problem Statement

The project will investigate whether supervised deep-learning models can classify individual, cropped peripheral blood smear WBC images into five categories: neutrophils, lymphocytes, monocytes, eosinophils, and basophils. The work will use a publicly available, labelled image dataset and will assess model performance, class-wise errors, and the feasibility of presenting predictions through a simple prototype.

3. Project Objectives

- Inspect and validate the dataset structure, labels, image formats, and class distribution.
- Perform EDA and visualize representative images from all five WBC classes.
- Develop a baseline CNN model for five-class image classification.
- Compare the baseline with at least one transfer-learning model, such as MobileNetV2 or ResNet50, subject to compute availability.
- Evaluate performance using accuracy, precision, recall, macro F1-score, and a confusion matrix.
- Examine misclassified examples and discuss class imbalance, dataset limitations, and generalizability.
- Build a lightweight Streamlit demonstration that accepts one cropped WBC image and returns a predicted class and model confidence.

4. Target Users and Use Case

The intended users are learners and researchers in medical image analysis, pathology, hematology, and AI/ML. The prototype will demonstrate image classification for educational and research purposes. It is not intended to replace a pathologist, provide a definitive diagnosis, or guide patient management.

5. Dataset Details

Field	Details
Dataset name	White Blood Cells Dataset
Source	Kaggle – Masoud Nickparvar
URL	https://www.kaggle.com/datasets/masoudnickparvar/white-blood-cells-dataset/data
Task	Five-class classification of individual WBC images
Classes	Basophil, eosinophil, lymphocyte, monocyte, neutrophil
Observed Test-A directory	The provided screenshot shows 4,339 images: basophil 89, eosinophil 322, lymphocyte 1,034, monocyte 234, neutrophil 2,660.
Data verification	The complete dataset size, folder structure, labels, split definitions, and any overlap between splits must be verified after access. The Test-A count should not be treated as the full dataset size.

6. Steps to Access and Prepare the Dataset

- Access the Kaggle dataset and review its description, licence/usage terms, directory structure, and labels.
- Download or load a manageable subset first and confirm that images can be read correctly.
- Create a dataframe containing image paths and class labels; inspect corrupt, duplicate, or ambiguous files where feasible.
- Visualize representative examples and calculate class frequencies.
- Resize images to a consistent input size and normalize pixel values according to the selected model.
- Use the dataset's existing splits when their purpose and integrity are clear; otherwise create stratified train/validation/test splits without leakage.
- Apply augmentation only to training data and document all preprocessing decisions.

7. Scope of Work

In Scope

- Classification of cropped WBC images into the five labelled classes.
- EDA, image preprocessing, and assessment of class imbalance.
- Training a baseline CNN and comparing it with a transfer-learning model.
- Performance evaluation, class-wise analysis, and error review.
- A small Streamlit image-upload prediction demo and reproducible project documentation.

Out of Scope

- Detection or localization of WBCs in whole-slide or full-field peripheral blood smear images.
- Automated differential WBC counting from an entire smear.
- Diagnosis of leukemia, infection, or other diseases from WBC images.
- Clinical deployment, clinical validation, or claims of replacing expert review.

8. Recommended Technical Approach

- Use Python with a suitable framework such as TensorFlow/Keras or PyTorch; choose one framework and document versions.
- Establish a reproducible workflow for data loading, preprocessing, splitting, training, and evaluation.
- Train a simple CNN baseline.
- Fine-tune a pretrained MobileNetV2 or ResNet50 model, selecting the architecture based on available computing resources.
- Address imbalance through appropriate class weights or training-only augmentation if justified by EDA.
- Record experiments, random seeds, hyperparameters, training curves, and model checkpoints.
- Compare models on the same held-out test set and inspect false predictions.
- Package the best model in a Streamlit interface; include setup instructions and example predictions.

9. Model Evaluation Plan

Report accuracy, precision, recall, and F1-score, including macro-averaged metrics and per-class results. Include a confusion matrix, learning curves, examples of correct and incorrect predictions, and inference time if feasible. Explain the effect of class imbalance and avoid relying on accuracy alone. Do not report performance figures until experiments have actually been run.

10. Deployment Expectation

- A Streamlit application that loads a saved model without retraining.
- An image-upload control with predicted WBC category and confidence score.
- At least three sample predictions and one known failure case.
- A README documenting environment setup, dependencies, model loading, usage, and limitations.

11. Project Work Plan

- Phase 1 – Dataset access, verification, and project framing.
- Phase 2 – EDA, preprocessing, and leakage-safe data preparation.
- Phase 3 – Baseline CNN implementation.
- Phase 4 – Transfer-learning experiment and tuning.
- Phase 5 – Evaluation, error analysis, and interpretation.
- Phase 6 – Streamlit prototype, documentation, and final presentation.

12. Expected Deliverables

- Approved problem statement and scope document.
- Dataset audit and EDA notebook.
- Preprocessing and reproducible data-splitting pipeline.
- Baseline CNN and transfer-learning notebooks/scripts.
- Evaluation report with metrics, plots, confusion matrix, and error analysis.
- Saved model artifact and Streamlit demonstration.
- README/project repository and final presentation.

13. Success Criteria

- Dataset access, class labels, and data splits are documented and reproducible.
- A baseline and an improved/comparative model are implemented.
- Evaluation uses appropriate aggregate and class-wise metrics on held-out data.
- The prototype produces predictions for previously unseen input images.
- Results, failure cases, limitations, and potential sources of bias are explained honestly.

14. Risks, Limitations, and Responsible Use

- Class imbalance may reduce sensitivity for rare classes; assess class-wise recall and macro F1-score.
- Image quality, staining, illumination, magnification, and source differences may affect performance.
- Potential duplicate images or related images across splits can inflate results; inspect for leakage where feasible.
- The dataset may not represent all laboratories, populations, or abnormal cells.
- The tool is an educational proof of concept and must not be used for clinical decisions without appropriate external validation and governance.

15. Research Reference

Tavakoli S, Ghaffari A, Mousavi Kouzehkanan Z. “Generalizability in White Blood Cells’ Classification Problem.”
Uploaded reference article supplied for project planning. Use it as background for CNN-based WBC classification and dataset generalizability; do not claim to reproduce its exact method unless that method is implemented and evaluated.

Note: This is a proposed project plan. Dataset details beyond those visible in the supplied Kaggle screenshots must be verified directly from the dataset before submission. Project number, mentor approval, final framework, and timeline can be updated as required.